

COMPARISON OF LIQUID METAL CANDIDATES

Name	Melting point	Stability	Toxicity
Mercury (Hg)	-38.83°C	The only metallic element that is liquid at standard conditions for temperature and pressure	Mercury and most of its compounds are extremely toxic
Cesium (Cs)	28.5°C	Highly explosive when it comes in contact with water	Violently reacts with water
Gallium (Ga)	29.8°C	Elemental gallium is not found in nature, but it can be easily obtained by smelting	Reactive metal
Rubidium (Rb)	39.3°C	Spontaneously ignites to form rubidium oxide	Violently reacts with water

also help produce materials that are extremely bendable, paving the way for flexible electronics.

Metals like gallium and indium have a low melting point. These produce an atomically thin layer of oxide on their surface that naturally protects them. It is this thin oxide which is used in fabrication method. By rolling the liquid metal, the oxide layer can be transferred onto an electronic wafer, which is then sulphurised. The surface of the wafer can be pre-treated to form individual transistors. This novel method has been used to create transistors and photo-detectors of very high gain and very high fabrication reliability in large scale.

It has long been a dream in the electronics industry to be able to write out electronics directly, as simply as printing a picture onto paper with an office printer. The good news is that the first-ever prototype of a liquid-metal printer has been fabricated. It took optimisation of a series of industrial design and technical issues such as working reliability, printing resolution, automatic control, human-machine interface design, software, hardware, and integration between software and hardware, to create this high-quality personal desktop liquid-metal printer.

This liquid-metal printer is an automatic, easy-to-use and low-cost personal electronics manufacturing tool with many possible applications. As engineers have already manipulated the movements and shape of a drop

of liquid metal by adjusting the acidity of its watery environment, shape-shifting electronic circuits that can self-assemble are inching closer. Further, static circuit boards may one day give way to dynamic, changeable ones that can pop up on demand.

Gallium and Indium have different melting points, yet when mixed together, these form an alloy that is liquid at room temperature. Such an alloy is termed as an eutectic alloy. Researchers found that if they applied a small voltage (less than one volt) in water to a blob of the eutectic alloy, they were able to reduce the surface tension significantly, resulting in the molten metal spreading and flattening out. When the voltage was removed, the high surface tension returned, and the blob once more resumed its spherical shape.

The amount of surface tension could also be varied depending upon the voltage, making the blob more or less viscous. In other words, the liquid metal could be held in various states of liquidity. This technique—a version of electrohydrodynamics in which electric currents affect the movement of liquid metals—relies on an oxide ‘skin’ forming on the surface of the alloy when a voltage is applied. This skin behaves as a surfactant, lowering the surface tension between the metal and the surrounding fluid. The significance of this behaviour is that the metal could be made to flow in and out of variously shaped capillaries or moulds, allowing the alloy to take different contours. If these shapes were that of an an-

tenna, for example, the metal could behave as a highly-variable or tunable antenna capable of morphing its shape to receive or transmit a wide range of different wavelengths all from the same component.

Researchers have experimented with other versions of shape-shifting antennae in the past, but this is the first time that they have utilised electricity—rather than mechanical deformation—to alter the shape. This technique to control the movement of liquid metals can be used to change the shape of antennae and complete or break circuits. It could also be used in microfluidic channels, MEMS, or photonic and optical devices. Many materials form surface oxides, so the work could extend beyond liquid metals.

Hurdles to overcome

Researchers have taken the first steps towards making the science fiction a reality by developing a way to control the surface tension of liquid metals with the application of very low voltages. This may offer opportunities in a new field of morphing electronic circuits, self-healing electronics or even self-assembling terminator-style robots. However, challenges remain before the potential of liquid-metal flexible printed electronics is leveraged for end-user electronic gadgets and more extensive applications in the future. These include significantly different adhesions of liquid metal inks on various substrates under different oxidation degrees, weakness of circuits, difficulty in fabricating high-accuracy devices, and low rate of good-quality production. **EFY**